MULTIPLE NUISANCE BREAKS AND SPURIOUS PREDICTABILITY UNDER PERSISTENT REGRESSORS

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Supplementary Material

This supplement gives the technical details behind the block-sieve empirical likelihood construction in the main manuscript. Section S1 records the finite-sample block projection identities using the Moore–Penrose inverse. Section S2 proves the stable-score decomposition and the conditional omitted-break failure theorem. Section S3 proves the known-partition Wilks theorem under projection-bias negligibility rather than full root-T undersmoothing. Section S4 gives the high-level fixed-order estimated-score likelihood transfer. Section S5 proves the shifted-array local-alternative calculation. Section S6 records implementation details for the reported simulations and the empirical application.

For a scalar array $\boldsymbol{a} = (a_1, \ldots, a_T)'$, write

$$\|\boldsymbol{a}\|_T^2 = T^{-1} \sum_{t=1}^{T} a_t^2, \qquad \|\boldsymbol{a}\|_\infty = \max_{t \le T} |a_t|.$$

Unless stated otherwise, $O_p(\cdot)$ and $o_p(\cdot)$ for growing vectors and matrices are in Euclidean and spectral norm, respectively.

S1 Block Projection Algebra

Fix a partition $\Lambda = (k_1, \ldots, k_q)$. Let $\boldsymbol{Q}_{K,\Lambda}$ and $\boldsymbol{M}_{K,\Lambda}$ be defined as in (3.1) and (3.2). Throughout this supplement, all projection equalities are equalities in the empirical Hilbert space $(\mathbb{R}^T, T^{-1}\boldsymbol{a}'\boldsymbol{b})$.

Lemma S1 (Block projection identities)**.** *For any partition* Λ*,*

$$\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{M}_{K,\Lambda} = \boldsymbol{0}, \qquad \boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{M}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}.$$

For every $\boldsymbol{c}$ *conformable with* $\boldsymbol{Q}_{K,\Lambda}$*,*

$$(\boldsymbol{Q}_{K,\Lambda}\boldsymbol{c})'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = 0.$$

Proof. Let $\boldsymbol{\Pi}_{K,\Lambda} = \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{\dagger}\boldsymbol{Q}'_{K,\Lambda}$. This is the orthogonal projector onto the column space of $\boldsymbol{Q}_{K,\Lambda}$. Therefore $\boldsymbol{\Pi}'_{K,\Lambda} = \boldsymbol{\Pi}_{K,\Lambda}$, $\boldsymbol{\Pi}^2_{K,\Lambda} = \boldsymbol{\Pi}_{K,\Lambda}$, and $\boldsymbol{Q}'_{K,\Lambda}(\boldsymbol{I}_T - \boldsymbol{\Pi}_{K,\Lambda}) = \boldsymbol{0}$. Since $\boldsymbol{M}_{K,\Lambda} = \boldsymbol{I}_T - \boldsymbol{\Pi}_{K,\Lambda}$, the displayed identities follow. The final equality is obtained by multiplying $\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{M}_{K,\Lambda} = \boldsymbol{0}$ by $\boldsymbol{c}$ and $\boldsymbol{w}$. □

S2 Stable-Score Failure

This section proves Proposition 1 and Theorem 1. Let $\Lambda = 0$ denote the no-break partition and write $\boldsymbol{M}_{K,0}$ for the corresponding stable-nuisance residual-maker.

Proof of Proposition 1. Under $H_0 : \beta = \beta_0$,

$$\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{\mu} + \boldsymbol{U}.$$

Multiplying by the stable residualized weight gives the exact identity

$$T^{-1/2}(\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag})' \boldsymbol{M}_{K,0} \boldsymbol{w} = T^{-1/2} \boldsymbol{U}' \boldsymbol{M}_{K,0} \boldsymbol{w} + T^{-1/2} \boldsymbol{\mu}' \boldsymbol{M}_{K,0} \boldsymbol{w}.$$

The second term is the omitted-break component. No stable-sieve approximation remainder is needed because B_T is defined using the exact nuisance vector $\boldsymbol{\mu}$. □

Proof of Theorem 1. The scalar EL expansion gives

$$\ell_T^{\text{st}}(\beta_0) = \frac{\{S_T^{\text{st}}(\beta_0)\}^2}{V_T^{\text{st}}} + o_p(1).$$

By Proposition 1, $S_T^{\text{st}}(\beta_0) = A_T^{\text{st}} + B_T$. With

$$a_T = \frac{A_T^{\text{st}}}{(V_T^{\text{st}})^{1/2}}, \qquad b_T = \frac{B_T}{(V_T^{\text{st}})^{1/2}},$$

the expansion is $\ell_T^{\text{st}}(\beta_0) = (a_T + b_T)^2 + o_p(1)$. If $a_T = O_p(1)$ and $|b_T| \to_p \infty$, then $|a_T + b_T| \geq |b_T| - |a_T| \to_p \infty$, so the statistic diverges in probability. Under joint stable convergence $(a_T, b_T) \xrightarrow{\text{st}} (Z, B)$, the continuous mapping theorem gives $(a_T + b_T)^2 \Rightarrow (Z + B)^2$. Since Z is conditionally standard normal given the limiting sigma-field, $(Z + B)^2 \mid B \sim \chi_1^2(B^2)$. □

Remark S1 (Scope of omitted-break divergence)**.** The divergence conclusion is conditional on the stable empirical-likelihood quadratic expansion.

With a fixed omitted nuisance break, B_T may be of order $\sqrt{T}$, and the null-style scalar expansion need not hold automatically. A primitive fixed-break divergence proof therefore requires a separate expansion under the misspecified stable nuisance projection.

S3 Known-Partition Wilks Theorem

Set $\Lambda = \Lambda_0$ and write

$$\boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}, \qquad \widehat{\boldsymbol{u}}_0(\beta_0) = \boldsymbol{M}_{K,\Lambda_0}(\boldsymbol{Y} - \beta_0\boldsymbol{X}_{lag}).$$

Under the true partition,

$$\boldsymbol{Y} - \beta_0\boldsymbol{X}_{lag} = \boldsymbol{Q}_{K,\Lambda_0}\boldsymbol{c}_0 + \boldsymbol{r}_0 + \boldsymbol{U},$$

where $\boldsymbol{r}_0$ stacks the regime-specific sieve approximation errors. Let $\boldsymbol{e}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{r}_0$.

Lemma S2 (Known-partition replacement)**.** *Under Assumptions 3 and 4,*

$$T^{-1/2}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0,\Lambda_0) = T^{-1/2}\sum_{t=1}^{T}U_t v_{t,0} + o_p(1), \tag{S3.1}$$

$$T^{-1}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0,\Lambda_0)^2 = T^{-1}\sum_{t=1}^{T}U_t^2 v_{t,0}^2 + o_p(1), \tag{S3.2}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0,\Lambda_0)| = o_p(\sqrt{T}). \tag{S3.3}$$

Proof. Using Lemma S1,

$$\widehat{\boldsymbol{u}}_0(\beta_0) = \boldsymbol{M}_{K,\Lambda_0}(\boldsymbol{U} + \boldsymbol{r}_0), \qquad \boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}.$$

Because $\boldsymbol{M}_{K,\Lambda_0}$ is symmetric and idempotent,

$$T^{-1/2}\widehat{\boldsymbol{u}}_0(\beta_0)'\boldsymbol{v}_0 = T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_0)'\boldsymbol{v}_0.$$

By Assumption 3,

$$T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0 = T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} = o_p(1),$$

which proves (S3.1).

It remains to record the denominator and maximum bounds. Let $G_T = T^{-1}\boldsymbol{Q}_{K,\Lambda_0}'\boldsymbol{Q}_{K,\Lambda_0}$. Since the rows of $\boldsymbol{Q}_{K,\Lambda_0}$ have norm at most ζ_K and the eigenvalues of G_T are bounded away from zero and infinity,

$$\|\boldsymbol{v}_0\|_\infty = O_p(1 + \zeta_K).$$

Moreover,

$$T^{-1}\boldsymbol{Q}_{K,\Lambda_0}'\boldsymbol{r}_0 = O_p(a_K),$$

and hence

$$\|\boldsymbol{e}_0\|_T \leq \|\boldsymbol{r}_0\|_T = O_p(a_K), \qquad \|\boldsymbol{e}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

The stated rates imply

$$\|\boldsymbol{e}_0 \odot \boldsymbol{v}_0\|_T = O_p\{a_K(1+\zeta_K)\} = o_p(1), \qquad \|\boldsymbol{e}_0 \odot \boldsymbol{v}_0\|_\infty = O_p\{a_K(1+\zeta_K)^2\} = o_p(\sqrt{T}).$$

For the denominator, write $\widetilde{\boldsymbol{U}}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{U}$. Then

$$\widehat{Z}_t(\beta_0, \Lambda_0) = \widetilde{U}_{0,t}v_{t,0} + e_{0,t}v_{t,0}.$$

Assumption 4 gives

$$T^{-1}\sum_{t=1}^{T}\{(\widetilde{U}_{0,t}v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1),$$

and Cauchy–Schwarz with $\|\boldsymbol{e}_0 \odot \boldsymbol{v}_0\|_T = o_p(1)$ gives the remaining cross and square terms. This proves (S3.2). The maximum bound follows from

$$|\widehat{Z}_t(\beta_0, \Lambda_0)| \le |U_t v_{t,0}| + |(\widetilde{U}_{0,t} - U_t)v_{t,0}| + |e_{0,t}v_{t,0}|,$$

Assumption 4, and $\|\boldsymbol{e}_0 \odot \boldsymbol{v}_0\|_\infty = o_p(\sqrt{T})$. □

Lemma S3 (EL expansion)**.** *Let $Z_{t,T}$ be scalar scores and $\widehat{V}_T^Z = T^{-1}\sum_t Z_{t,T}^2$. Suppose zero lies in the interior of the scalar convex hull with probability tending to one and*

$$T^{-1/2}\sum_t Z_{t,T} = O_p(1), \qquad \widehat{V}_T^Z \to_p \eta^2, \qquad \max_t |Z_{t,T}| = o_p(\sqrt{T}),$$

where $P(\eta^2 \ge c) = 1$ for some fixed $c > 0$. Then

$$2\sum_{t=1}^{T}\log(1 + \widehat{\lambda}Z_{t,T}) = \frac{\{T^{-1/2}\sum_t Z_{t,T}\}^2}{T^{-1}\sum_t Z_{t,T}^2} + o_p(1).$$

Proof. The second-moment condition implies $P(\widehat{V}_T^Z \ge c/2) \to 1$. Let $S_T^Z = \sum_t Z_{t,T}$ and $Q_T^Z = \sum_t Z_{t,T}^2$. On this event, $S_T^Z = O_p(\sqrt{T})$ and $Q_T^Z \ge cT/2$.

Monotonicity of the scalar multiplier equation, evaluated at $\pm 2|S_T^Z|/Q_T^Z$, gives

$$|\widehat{\lambda}| \leq 2|S_T^Z|/Q_T^Z = O_p(T^{-1/2}).$$

Since $\max_t |\widehat{\lambda} Z_{t,T}| = o_p(1)$, a Taylor expansion of the multiplier equation gives

$$\widehat{\lambda} = \frac{\sum_t Z_{t,T}}{\sum_t Z_{t,T}^2} + o_p(T^{-1/2}).$$

Substitution into the second-order expansion of the log empirical likelihood gives the displayed quadratic approximation. □

Proof of Theorem 2. Apply Lemma S2 to reduce the feasible score to the matched block-projection oracle score $U_t v_{t,0}$. Assumption 4 gives the stable numerator limit, denominator consistency, feasible convex hull, and maximum-score condition. Lemma S3 then yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{T^{-1/2} \sum_t U_t v_{t,0}\}^2}{T^{-1} \sum_t U_t^2 v_{t,0}^2} + o_p(1) \Rightarrow \chi_1^2.$$

□

S4 Estimated-Partition Equivalence

This section records the high-level likelihood-transfer argument behind Theorem 3. Let

$$S_T(\Lambda) = T^{-1/2}\sum_t \widehat{Z}_t(\beta_0,\Lambda), \qquad V_T(\Lambda) = T^{-1}\sum_t \widehat{Z}_t(\beta_0,\Lambda)^2.$$

Lemma S4 (Fixed-order estimated-partition likelihood transfer)**.** *Under the assumptions of Theorem 2 and Assumption 6,*

$$\ell_T(\beta_0,\widetilde{\Lambda}) - \ell_T(\beta_0,\Lambda_0) = o_p(1).$$

Proof. Lemma S2 and Assumption 4 give $S_T(\Lambda_0) = O_p(1)$, $V_T(\Lambda_0) \to_p \eta_q^2$, and a true-partition maximum-score bound. Assumption 6 gives

$$S_T(\widetilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \qquad V_T(\widetilde{\Lambda}) - V_T(\Lambda_0) = o_p(1),$$

and the fixed-order estimated-partition maximum-score and convex-hull conditions. Lemma S3 applies to both arrays, and the two quadratic ratios differ by $o_p(1)$ because both denominators are bounded away from zero with probability tending to one. □

Proof of Theorem 3. By Lemma S4,

$$\ell_T(\beta_0,\widetilde{\Lambda}) = \ell_T(\beta_0,\Lambda_0) + o_p(1).$$

The result follows from Theorem 2. If $P\{\widehat{q}(\beta_0) = q_0\} \to 1$, then $\widehat{\Lambda}(\beta_0) = \widetilde{\Lambda}$

with probability tending to one. This proves Corollary 1 under the fixed-order event version of Assumption 6. □

S5 Local Predictive Alternatives

This section proves Theorem 4. Under the local alternative $\beta_T = \beta_0 + d_T$, the null-imposed residual satisfies

$$\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{\mu} + \boldsymbol{U} + d_T \boldsymbol{X}_{lag}.$$

At the true partition, the block projection removes the regime-specific sieve approximation to $\boldsymbol{\mu}$. The numerator replacement condition in Theorem 4 gives

$$T^{-1/2} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^{T} U_t v_{t,0} + T^{-1/2} \sum_{t=1}^{T} g_{t,T} + o_p(1).$$

By the local drift condition, the second term converges to $\eta_q \delta_h$. The denominator replacement condition reduces the feasible denominator to the shifted oracle denominator. Moreover,

$$T^{-1} \sum_t (U_t v_{t,0} + g_{t,T})^2 = V_T^{(q)} + 2T^{-1} \sum_t U_t v_{t,0} g_{t,T} + T^{-1} \sum_t g_{t,T}^2.$$

The last term is $o_p(1)$ by the local square condition, and the cross term is $o_p(1)$ by Cauchy–Schwarz and $V_T^{(q)} = O_p(1)$. Thus the feasible denominator remains $\eta_q^2 + o_p(1)$. The maximum condition follows from the oracle maximum condition and the local maximum and replacement conditions. The feasible convex-hull condition and Lemma S3 give the known-partition limit $(Z + \delta_h)^2$. Lemma S4, applied under the local-alternative estimated-score and convex-hull conditions, transfers the result to $\widetilde{\Lambda}$. Correct order selection gives the unknown-order statement for $\ell_T^{BA}(\beta_0)$.

S6 Implementation Details

For simulations, each replication uses the same null-imposed profile path as the theory:

$$\beta \to \widehat{q}(\beta), \widehat{\Lambda}(\beta) \to \boldsymbol{M}_{K,\widehat{\Lambda}(\beta)} \to \widehat{Z}_t(\beta) \to \ell_T^{BA}(\beta).$$

The proof object is global profile-sieve segmentation. Computation may use multiscale candidate generation, followed by final selection with the profile-sieve RSS and the information criterion in (4.4). The implementation must report the candidate-generation method separately from the statistical estimator, because SBS, NOT, and related devices are computational accelerators rather than the main contribution.

For empirical work, the stable and break-adaptive statistics are reported together. The stable statistic documents conventional evidence of predictability; the adaptive statistic documents whether that evidence survives multiple nuisance breaks. The empirical section also reports $\widehat{q}$, break dates, regime-specific nuisance fits, an estimated break-induced score component, and residualized predictor strength $T^{-1}\boldsymbol{w}'\boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}$.